

Target Table 1

Overview

The target table contains Cyclistic trip data used for dashboard visualizations. The full table includes about 12.5 million rows. The sample extract (1000 rows) is provided as a separate file: **Cyclistic_target_table_sample.csv**

Pipeline code:

```
SELECT
    TRI.usertype,
    TRI.start_station_id,
    TRI.start_station_name,
    ZIPSTART.zip_code AS zip_code_start,
    ZIPSTARTNAME.borough borough_start,
    ZIPSTARTNAME.neighborhood AS neighborhood_start,
    TRI.end_station_name,
    ZIPEND.zip_code AS zip_code_end,
    ZIPENDNAME.borough borough_end,
    ZIPENDNAME.neighborhood AS neighborhood_end,
    DATE(TRI.starttime) AS start_day,
    DATE(TRI.stoptime) AS stop_day,
    WEA.temp AS day_mean_temperature, -- Mean temp
    WEA.wdsp AS day_mean_wind_speed, -- Mean wind speed
    WEA.prcp AS day_total_precipitation, -- Total precipitation
    -- Group trips into 10 minute intervals to reduces the number of rows
    ROUND(CAST(TRI.tripduration / 60 AS INT64), -1) AS trip_minutes,
    COUNT(TRI.bikeid) AS trip_count
FROM
    `bigquery-public-data.new_york_citibike.citibike_trips` AS TRI
INNER JOIN
    `bigquery-public-data.geo_us_boundaries.zip_codes` ZIPSTART
ON ST_WITHIN(
    ST_GEOPOINT(TRI.start_station_longitude, TRI.start_station_latitude),
```

```

ZIPSTART.zip_code_geom)
INNER JOIN
  `bigquery-public-data.geo_us_boundaries.zip_codes` ZIPEND
ON ST_WITHIN(
  ST_GEOGPOINT(TRI.end_station_longitude, TRI.end_station_latitude),
  ZIPEND.zip_code_geom)
INNER JOIN
  `bigquery-public-data.noaa_gsod.gsod20*` AS WEA
ON  PARSE_DATE("%Y%m%d",  CONCAT(WEA.year,  WEA.mo,  WEA.da))  =
DATE(TRI.starttime)
INNER JOIN
  `my-bi-project-23653.cyclistic_project.cyclistic_zip` AS ZIPSTARTNAME
ON ZIPSTART.zip_code = CAST(ZIPSTARTNAME.zip AS STRING)
INNER JOIN
  `my-bi-project-23653.cyclistic_project.cyclistic_zip` AS ZIPENDNAME
ON ZIPEND.zip_code = CAST(ZIPENDNAME.zip AS STRING)
WHERE
  -- This takes the weather data from one weather station
  WEA.wban = '94728' -- NEW YORK CENTRAL PARK
  -- Use data from 2014 and 2015
AND
  EXTRACT(YEAR FROM DATE(TRI.starttime)) BETWEEN 2014 AND 2015
GROUP BY
  1,
  2,
  3,
  4,
  5,
  6,
  7,

```

8,
9,
10,
11,
12,
13,
14,
15,
16

Base table adapted from course template

Table schema:

Field Name	Type	Description
usertype	VARCHAR	Customer / Subscriber (annual member)
start_station_id	INTEGER	ID of the start station
start_station_name	VARCHAR	Start station name
zip_code_start	INTEGER	Zip code area of the start station
neighborhood_start	VARCHAR	Neighborhood of the start station
borough_start	VARCHAR	Borough of the start station
end_station_name	VARCHAR	End station name
zip_code_end	INTEGER	Zip code area of the end station
neighborhood_end	VARCHAR	Neighborhood of the end station
borough_end	VARCHAR	Borough of the end station
start_day	DATE	Trip start date
stop_day	DATE	Trip end date
day_mean_temperature	FLOAT	Daily average temperature
day_mean_wind_speed	FLOAT	Daily average wind speed
day_total_precipitation	FLOAT	Daily total precipitation
trip_minutes	INTEGER	Trip duration (rounded to nearest 10 minutes)
trip_count	INTEGER	Aggregated number of trips

Target Table 2

Overview

The target table contains net flow numbers for each Cyclistic bike station. The sample extract (1000 rows) is provided as a separate file: **Cyclistic_congestion_table_sample.csv**

Pipeline code:

```
SELECT
    s.*,
    e.end_trips,
    e.end_trips - s.start_trips AS net_flow,
    d.neighborhood,
    d.borough
FROM (
    SELECT
        start_station_id AS station_id,
        start_station_name AS station_name,
        start_station_latitude AS station_latitude,
        start_station_longitude AS station_longitude,
        DATE(starttime) AS trip_date,
        COUNT(start_station_id) AS start_trips
    FROM
        bigquery-public-data.new_york_citibike.citibike_trips
    GROUP BY
        1,2,3,4,5
) AS s
INNER JOIN (
    SELECT
        end_station_id AS station_id,
        end_station_name AS station_name,
        DATE(starttime) AS trip_date,
```

```

COUNT(end_station_id) AS end_trips
FROM
bigquery-public-data.new_york_citibike.citibike_trips
GROUP BY
1,2,3
) AS e
ON
s.station_id = e.station_id
AND
s.trip_date = e.trip_date
--I created dim_stations table using Target table 1 to speed up joining
process. dim_stations table contains the following fields: station_id,
station_name, station_neighborhood, station_borough
INNER JOIN
my-bi-project-23653.cyclistic_project.dim_stations AS d
ON
s.station_id = d.station_id
WHERE
s.trip_date BETWEEN '2014-01-01' AND '2015-12-31'

```

Field Name	Type	Description
station_id	INTEGER	ID of the station
station_name	VARCHAR	Station name (identical for start and end)
station_latitude	FLOAT	Station latitude (identical for start and end)
station_longitude	FLOAT	Station longitude (identical for start and end)
trip_date	DATE	Date of the trips
start_trips	INTEGER	The amount of start trips
end_trips	INTEGER	The amount of end trips
net_flow	INTEGER	The amount of net flow
neighborhood	VARCHAR	Neighborhood of the station
borough	VARCHAR	Borough of the station

Target Table 3

Overview

The target table contains Cyclistic trip data with full date-time information from July to September 2015 used for daily peak usage time analysis. The full table includes about 3.5 million rows. The sample extract (1000 rows) is provided as a separate file: **Cyclistic_daytime_table_sample.csv**

Pipeline code:

```
SELECT

    trip.usertype,

    trip.start_station_id,

    trip.start_station_name,

    trip.starttime,

    COUNT(*) AS trip_count

FROM

    bigquery-public-data.new_york_citibike.citibike_trips AS trip

WHERE

    trip.starttime BETWEEN '2015-07-01' AND '2015-09-30'

GROUP BY

    1,

    2,

    3,

    4
```

Table schema:

Field Name	Type	Description
usertype	VARCHAR	Customer / Subscriber (annual member)
start_station_id	INTEGER	ID of the start station
start_station_name	VARCHAR	Start station name
starttime	TIMESTAMP	Trip start date and time
trip_count	INTEGER	Number of trips